

Visual Inspection Criteria for Finished Products

Purpose

This work instruction defines the visual acceptance criteria for finished products at Acme Corporation.

Scope

This instruction applies to all finished product visual inspections performed by QA inspectors and production operators.

Inspection Conditions

All visual inspections must be performed under the following conditions:

- Lighting: 1000-1500 lux at the inspection surface
- Distance: 30-40 cm from the inspector's eyes
- Angle: 15-45 degrees from normal
- Magnification: 2x for critical features, 1x for general inspection
- Environment: Clean, dust-controlled area
- Duration: Minimum 15 seconds per inspection surface

Defect Classification

Critical Defects

A critical defect renders the product unsafe or non-compliant with regulations. Zero critical defects are acceptable (AQL 0.0).

Examples: - Sharp edges or burrs that could cause injury - Missing safety labels - Incorrect material composition - Cracks in structural components - Leaks in sealed components

Major Defects

A major defect affects the form, fit, or function of the product. Maximum acceptable rate: 0.65% (AQL 0.65).

Examples: - Dimensional deviation exceeding ± 0.5 mm - Color variation exceeding 2 Delta E - Surface scratches deeper than 0.2 mm - Missing or incorrect components - Improper assembly - Cosmetic blemishes on visible surfaces - Incorrect labeling or packaging

Minor Defects

A minor defect does not affect form, fit, or function. Maximum acceptable rate: 2.5% (AQL 2.5).

Examples: - Slight color variation within 1-2 Delta E - Surface scratches less than 0.2 mm deep - Minor flash or burrs (removable with standard tool) - Slight cosmetic imperfections on non-visible surfaces

Inspection Procedure

1. Prepare the Inspection Station

- Clean the inspection surface
- Verify lighting meets specification
- Prepare inspection tools: ruler, go/no-go gauge, color standard
- Verify the product specification sheet is available

2. Inspect Each Surface

For each product, inspect in this order:

1. Top surface — rotate in 360 degrees
2. Bottom surface
3. Left side
4. Right side
5. Front
6. Back
7. Internal surfaces (if accessible)

3. Record Results

For each inspected product:

- Record the product serial number or batch ID
- Mark each defect found with a colored tag (red=critical, yellow=major, green=minor)
- Count and record the number of each defect type
- Record the inspector name and inspection time

4. Acceptance Decision

Condition	Decision
Any critical defect	Reject entire batch. Initiate NCR per SOP-QA-012
Major defects > AQL 0.65%	Reject batch. Sort 100%
Minor defects > AQL 2.5%	Reject batch. Sort or negotiate with QA Manager
Within AQL limits	Accept

Sampling Plan

Standard inspection level: II per ANSI/ASQ Z1.4.

For lots up to 500 pieces: Sample size 50. Accept on 5 major defects, reject on 6.

For lots 501-1200: Sample size 80. Accept on 7 major defects, reject on 8.

For lots over 1200: Sample size 125. Accept on 10 major defects, reject on 11.

Related Documents

- SOP-QA-012: Non-Conformance Report Procedure
- WI-055: First Article Inspection Procedure
- ANSI/ASQ Z1.4 Sampling Procedures

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